

Working with Concept Sets in Capr

Contents

0.1	Defining a concept set	1
0.2	Including Descendants	1
0.3	Toggling other logic	2
0.4	Fill in missing concept set details	2

library(Capr)

In version 2 of Capr, we have introduced a whole new way of building concept sets. Some highlights of this new system are, we no longer need a database connection, only need OMOP concept ids, and we provide functionality to improve how we map to the vocabulary (i.e. find descendants or exclude). The new interface makes it much easier to build and read information about a concept set. In this vignette we will demonstrate how the new AI works!

0.1 Defining a concept set

The new function `cs` is a binder function that collects integer OMOP ids into a set. The idea is for the function to look like `c()` which in R is use to formulate vectors. Say we found several ingredients that are all ace inhibitors. We can combine those OMOP concept ids into a set as follows:

```
ace1 <- cs(1335471, #benazepril
          1340128, #captopril
          1341927, #enalapril
          1308216, #lisinopril
          1363749, #fosinopril
          name = "aceInhibitors")

ace1
#> -- <Capr Concept Set> aceInhibitors -----
#> # A tibble: 5 x 9
#>   conceptId conceptCode conceptName domainId vocabularyId standardConcept includeDescendants isExclu
#>   <int>    <chr>        <chr>        <chr>    <chr>          <chr>        <lgl>          <lgl>
#> 1  1335471 ""          ""          ""         ""            ""          FALSE         FALSE
#> 2  1340128 ""          ""          ""         ""            ""          FALSE         FALSE
#> 3  1341927 ""          ""          ""         ""            ""          FALSE         FALSE
#> 4  1308216 ""          ""          ""         ""            ""          FALSE         FALSE
#> 5  1363749 ""          ""          ""         ""            ""          FALSE         FALSE
```

0.2 Including Descendants

If you were only add the OMOP id for the ingredient, our query would only look at whether that specific ID is among the drug concepts for a set of patients. With OMOP, we often don't want just the ingredient ID but also its descendants. Descendants are child concepts that map to the adult concepts. For example a brand name or specific dosage of lisinopril would have a different OMOP concept Id but it would map to the ingredient term via the vocabulary hierarchy. This helps us include all variations of linisopril that could be seen in the data. In Capr we can easily add this logic using the `descendants` command

```
ace2 <- cs(descendants(1335471, 1340128, 1341927, 1363749, 1308216),
           name = "aceInhibitors")
ace2
#> -- <Capr Concept Set> aceInhibitors -----
#> # A tibble: 5 x 9
#>   conceptId conceptCode conceptName domainId vocabularyId standardConcept includeDescendants isExclu
#>   <int>    <chr>        <chr>        <chr>    <chr>          <chr>          <lgl>          <lgl>
#> 1   1335471 ""          ""          ""        ""            ""            TRUE          FALSE
#> 2   1340128 ""          ""          ""        ""            ""            TRUE          FALSE
#> 3   1341927 ""          ""          ""        ""            ""            TRUE          FALSE
#> 4   1363749 ""          ""          ""        ""            ""            TRUE          FALSE
#> 5   1308216 ""          ""          ""        ""            ""            TRUE          FALSE
```

0.3 Toggling other logic

A similar strategy can be used to exclude concepts from a set or include mapped. We can also combine this within the `cs` function, as shown below

```
ace3 <- cs(descendants(1335471, 1340128, 1341927, 1363749), exclude(1308216),
           name = "aceInhibitors")
ace3
#> -- <Capr Concept Set> aceInhibitors -----
#> # A tibble: 5 x 9
#>   conceptId conceptCode conceptName domainId vocabularyId standardConcept includeDescendants isExclu
#>   <int>    <chr>        <chr>        <chr>    <chr>          <chr>          <lgl>          <lgl>
#> 1   1335471 ""          ""          ""        ""            ""            TRUE          FALSE
#> 2   1340128 ""          ""          ""        ""            ""            TRUE          FALSE
#> 3   1341927 ""          ""          ""        ""            ""            TRUE          FALSE
#> 4   1363749 ""          ""          ""        ""            ""            TRUE          FALSE
#> 5   1308216 ""          ""          ""        ""            ""            FALSE         TRUE
```

0.4 Fill in missing concept set details

By default, `Capr` puts in minimal information for the concept set, the concept id. However it is often helpful to fill out the remaining information of the concept using the OMOP vocabularies. To fill out the remaining information, a user needs to connect to a OMOP CDM to access the vocabulary information for the concept in the concept set. First we show a regular concept set, and the json it renders.

```
diclofenac <- cs(descendants(1124300), name = "diclofenac")
cat(as.json(diclofenac))
```

Now we add in the CDM connection to get the remaining information:

```
con <- DatabaseConnector::connect(Eunomia::getEunomiaConnectionDetails())
diclofenac <- getConceptSetDetails(diclofenac, con, vocabularyDatabaseSchema = "main")
cat(as.json(diclofenac))
```